

Industrial Error Code Reference

Synthetic project reference document for Phase 5 RAG testing.

1. Purpose

This project reference maps representative industrial error codes to symptoms, possible causes, and recommended investigation steps. Codes are illustrative and must not be treated as manufacturer-specific codes unless confirmed against actual equipment documentation.

2. Error Codes

E101 — Motor Overtemperature: possible overload, poor cooling, bearing friction, or electrical fault. E102 — Excessive Vibration: possible imbalance, misalignment, bearing wear, looseness, or tool imbalance. E103 — Low Hydraulic Pressure: possible leak, low fluid, clogged filter, pump wear, or valve issue.

3. Additional Error Codes

E104 — Spindle Vibration: possible spindle bearing degradation, tool imbalance, alignment problem, or looseness. E105 — Motor Overcurrent: possible overload, mechanical resistance, bearing degradation, or electrical fault. E106 — RPM Instability: possible drive issue, overload, controller problem, coupling problem, or changing process load.

4. Additional Error Codes

E107 — Cooling System Warning: possible low coolant, blocked flow, pump problem, or contamination. E108 — Lubrication Warning: possible low lubricant, blocked line, incorrect lubricant, or leakage. E109 — Bearing Temperature Warning: possible bearing wear, insufficient lubrication, overload, or misalignment. E110 — Maintenance Due: scheduled interval or condition-monitoring threshold reached.

5. Using Error Codes with Sensor Data

An error code should be interpreted together with current sensor data and machine history. For example, E104 combined with increasing vibration and bearing temperature provides stronger evidence for spindle-related investigation than E104 alone.

Important: This is synthetic reference content created for software testing and demonstration. It is not a manufacturer manual or a substitute for approved maintenance procedures, qualified personnel, or applicable safety requirements.